

Lab 1 – Knowledge Is Power (KIP) Description Paper

Rachel Thomas

Course: CS 411W

Instructor: Thomas J. Kennedy

Submission Date: September 29, 2026

Version 1

1. Introduction.....	2
1.1 Societal Problem.....	2
1.2 Needed Solution Characteristics.....	2
1.3 Product Introduction.....	3
2. KIP (Knowledge Is Power) Product Description.....	3
2.1. Key Product Features and Capabilities.....	4
2.1.3.1. "Fact or Fiction?" Game.....	4
2.1.3.2. "Choose the Outcome" Game.....	4
2.1.4. Progress Tracking and Analytics.....	5
2.2. Major Components (Hardware/Software).....	5
2.2.1. Presentation Layer.....	5
2.2.2. Application Layer.....	6
2.2.3. Data Layer.....	8
2.2.4. Hardware Layer.....	9
4. Glossary.....	11
5. References.....	12
Listing of Figures.....	13
Figure 1: KIP Major Functional Component Diagram (MFCD).....	13
Listing of Tables.....	14
Table 1: Presentation Layer.....	14
Table 2: Application Layer.....	15
Table 3: Data Layer.....	15
Table 4: Hardware Layer.....	16

1. Introduction

Knowledge Is Power (KIP) is a web-based educational platform designed to provide young adults with an accessible framework for legal literacy. The primary objective of the platform is to foster critical discernment, proactive information-seeking behavior, and personal accountability regarding everyday legal frameworks. By translating complex statutory language into structured, digestible learning formats, the KIP prototype demonstrates a scalable methodology for delivering independent legal education free from classroom distractions.

1.1 Societal Problem

Modern young adults interact with legally impactful environments at an accelerated rate, yet they lack formal training to navigate them safely. This issue is compounded by the proliferation of unverified, oversimplified legal content on digital social media platforms, which frequently generates inaccurate assumptions regarding statutory boundaries. This saturation of fragmented information creates a critical societal need for a centralized, objective service that evaluates an individual's specific legal comprehension. Without a reliable mechanism to assess baseline knowledge and identify cognitive gaps, young adults remain vulnerable to misinterpreting legal responsibilities, leading to avoidable civil and criminal liabilities.

1.2 Needed Solution Characteristics

To address this education gap, the platform must teach legal concepts through engaging, interactive activities that require users to apply laws to realistic situations rather than simply memorize abstract statutory definitions. To remain accessible to its target demographic, the

service should incorporate consistent language, concise legal explanations, mobile-friendly responsive web design, and low-stakes practice games that students can navigate independently. Additionally, the platform must explicitly demonstrate the connection between each lesson and its official legal sources, while introducing legal vocabulary directly in context so that users can recognize practical terms used across diverse legal topics.

1.3 Product Introduction

The Knowledge Is Power (KIP) platform serves as a supplemental educational tool engineered to mitigate youth legal illiteracy. KIP delivers an asynchronous, interactive environment that systematically guides users through localized legal frameworks and scenario-based evaluations. The primary objective of the core prototype is to demonstrate the operational feasibility of this educational delivery model through an integrated, web-accessible system. To validate its design requirements, the prototype instantiates a secure user authentication protocol, a progressive learning module architecture, and interactive branching scenario trees. Furthermore, the system incorporates a centralized data schema for progress tracking and a lightweight performance analytics engine to deliver targeted educational recommendations based on individual user metrics.

2. KIP (Knowledge Is Power) Product Description

The core objective of the solution is to assess a user's current baseline of legal knowledge, identify systemic misconceptions, and reinforce informed decision-making through repeated, low-stakes practice. If KIP is able to accurately capture the user's strengths and

weaknesses, the personalized curriculum will maintain engagement and foster autonomous learning.

2.1. Key Product Features and Capabilities

KIP empowers tech-friendly youth to navigate everyday legal boundaries confidently at their own time and pace, in hopes of cultivating long-term retention of their civil rights and responsibilities. If the KIP system can move from a position of official authority, to more of an omnipotent narrator of choices, curriculum will evolve into simulation of real-life consequences.

2.1.3.1. "Fact or Fiction?" Game

This short-form game presents users with common legal beliefs and widespread misconceptions, offering “true”, “false”, or “context-dependent” as selectable answers. This approach helps users challenge underlying assumptions, spot legal nuance, and build the foundational vocabulary necessary for more complex material. Crisp, meme-adjacent fair-use sound clips will lend itself to contemporary lightness.

2.1.3.2. "Choose the Outcome" Game

This multiple-choice, long-form game encourages users to fully interact with digital people, objects, and legal documents to investigate scenario details before rendering a decision. Every choice dynamically unfolds to demonstrate its immediate consequences to illuminate its real-world legal significance, training users to evaluate context and identify variables an outcome depends on.

2.1.4. Progress Tracking and Analytics

KIP continuously tracks topical strengths, weaknesses, and module completion rates during system interaction. This granular data allows KIP to algorithmically recommend relevant games and modules while optimizing active game sessions to deliver questions, scenarios, and legal concepts that meaningfully challenge the user's weak points.

2.2. Major Components (Hardware/Software)

The architecture of KIP follows a standard multi-tier software system structure modeled after the project's Major Functional Component Diagram (MFCD). It isolates user interaction, business logic, data storage, and underlying hardware infrastructure into distinct, modular layers.

2.2.1. Presentation Layer

The delivery methodology of the Knowledge Is Power (KIP) user interface is critical to its educational efficacy. Given the vast volume of digital information available to the target demographic, KIP deliberately rejects a traditional, top-down lecturing approach. Instead, the presentation layer is designed to act as an objective, peer-level guide that simplifies the complex statutory rules and regulations associated with modern adulthood. By prioritizing an approachable, intuitive, and non-authoritarian user experience, the interface lowers cognitive barriers to legal literacy and encourages autonomous, student-driven exploration.

2.2.1.1. Tech Stack: The frontend leverages a reliable network of established, modern technologies that will ensure future iterations will be maintainable and deliverable. KIP will

employ Angular / React, TypeScript, Bootstrap, HTML5, and Webpack. These tools are accelerated in our use of JHipster to support rapid and seamless deployment.

2.2.1.2. Account Portal: This system exposes UI components generated from JHipster's default user account management routines. It allows established users to securely authenticate and view progress metrics, while giving newcomers a clear entry path to utilize the platform under a temporary guest profile with limited local-session state tracking.

2.2.1.3. Module Browser: A responsive interface dashboard built around reusable client components designed by ingenious teammate Benjamin Keckley. It maps to underlying resource APIs to cleanly display interactive legal learning domains (e.g., traffic law, online conduct, school/work rights, personal safety), displaying the real-time progress.

2.2.1.4. Game Browser: A personalized client view that lists active application activities. It hooks into the backend's algorithmic filters to dynamically sort and display game recommendations tailored to the user's specific skill mastery profile.

2.2.1.5. Scenario Player: The dedicated interactive client interface engine that handles game rendering. It parses json-structured data models sent down from the server to smoothly present sequential evaluation screens, multi-option decision forks, contextual legal vocabulary tooltips, and real-time validation feedback loops.

2.2.2. Application Layer

The Application Layer serves as the core operational brain of the KIP platform, managing business logic processing, secure data translation, and game state workflows. To cultivate

long-term user adoption, the system's architecture must project absolute objectivity and reliability; any perceived inconsistency in data processing compromises user trust. This layer enforces that trust by serving as an objective, secure backbone that receives frontend client requests, evaluates programmatic learning rules, and interfaces directly with the MySQL database persistence tier.

2.2.2.1. Tech Stack

To ensure future iterations remain robust, secure, and easily deliverable, the backend architecture relies on a network of established and current enterprise tools. KIP exploits Java's type-safe architecture through the enterprise-grade tools provided by JHipster to guarantee a stable and highly maintainable codebase. Through this framework, KIP utilizes standardized REST APIs and Angular integration to decouple frontend and backend operations. This separation of concerns allows a clean, parallel development workflow that simplifies future system maintainability.

2.2.2.2. Authentication System: An access control framework configured via Spring Security utilizing stateless JSON Web Tokens (JWT). It manages secure user onboarding, session tokens, and automated authorization validation parameters, verifying whether incoming REST API calls are associated with a fully registered profile or a restricted guest account.

2.2.2.3. Scenario Engine: Dynamically loads and orchestrates the interactive content used during gameplay. For any active session, it assembles a "match" composed of multiple linear or branching "rounds," tracking unique choices, structural branches, consequences, and game states based on the specific game selected and user actions.

2.2.2.4. Feedback System: Generates precise structural evaluation objects passed back through the REST controller layer immediately following a user input. It isolates the decision parameters, evaluates if an entry is true, false, or context-dependent, and appends conversational, plain-language legal reasoning to explain why a selection was secure, hazardous, or incomplete.

2.2.2.5. Curation System: Executes data filtering algorithms to query the underlying repository layer based on identified user profile statistics. It systematically builds collection response bodies to prioritize weak legal categories while maintaining accessible, unrestricted catalog search capabilities.

2.2.2.6. Source Reference System: Bridges the educational content with structural reality by linking game outcomes to verified legal sources, legislative text, and relevant legal vocabulary. It tracks updates to active external legislation, updates relevant links automatically, and flags outdated KIP content for administrative review or temporary disabling.

2.2.3. Data Layer

The data persistence layer ensures secure, lightweight, and structured storage for all system assets, path-state relationships, and user profiles, managed via JHipster-generated Liquibase documents.

2.2.3.1. Tech Stack: PostgreSQL (Production) / H2 (Development/Testing), Spring Data JPA, Liquibase.

2.2.3.2. Databases & JDL Entities: Core relational database schema containing the generated JHipster User entity, alongside domain entities specified via JHipster Domain

Language (JDL): Progress, LearningModule, GameScenario, ScenarioNodeTransition, and LegalResource

2.2.3.3. Functions: Securely persists encrypted user account credentials and system authorities using standard JHipster security models; tracks quantitative user progress logs and dynamic path performance analytics; archives formal linear learning modules; stores game scenario metadata, sequential branching transition tables, and finite state node-tracking values using strict JPA ManyToOne self-referencing and OneToMany relational mappings; maps logical decision forks to precise legal outcomes; links structural legal citations to external educational resources.

2.2.4. Hardware Layer

The hardware topology supports the web-facing distribution, runtime evaluation, and containerized scaling of the platform components using a microservices-ready deployment architecture.

2.2.4.1. Server-Side Components: Server-Side Components: A secure Virtual Private Server (VPS) cluster or managed cloud environment running containerized service pods orchestrated via Docker Compose. This environment hosts the primary Spring Boot application image alongside its required service containers—specifically utilizing the generated `src/main/docker/postgresql.yml` configuration file to instantiate the isolated production database instance—and integrates standard JHipster toolings for registry tracking and environment management.

2.2.4.2. Client-Side Components: Standard, web-accessible consumer devices (smartphones, tablets, laptops, and desktop computers) utilizing a modern, standards-compliant web browser and active internet connection.

2.2.4.3. Functions: Executes containerized backend Java/Spring Boot application logic; manages database state transitions inside isolated volume mounts; scales application instances horizontally using Docker commands (docker compose scale) to handle variable traffic loads; reads, parses, and processes incoming API user requests via a centralized API Gateway; and pushes the responsive, state-tracking Angular/React web interface down to the client viewport.

4. Glossary

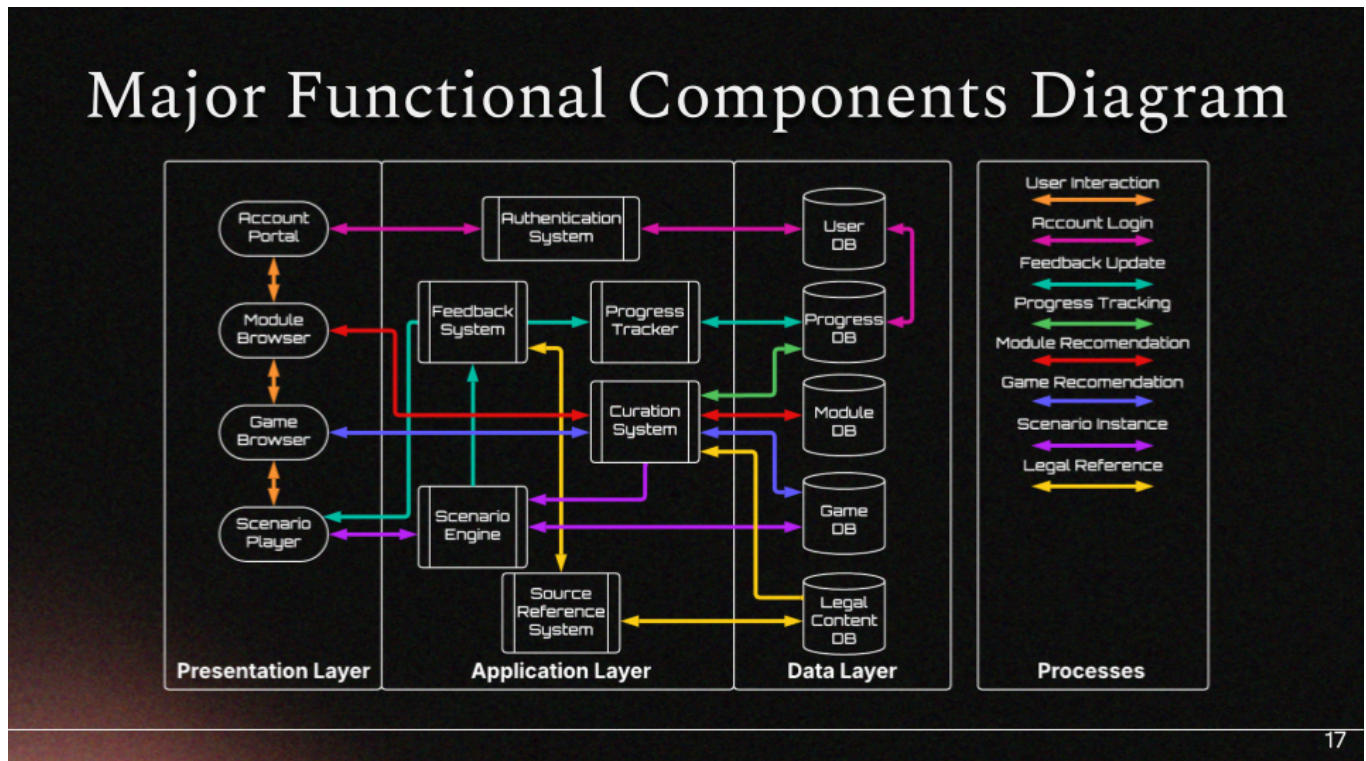
- API: Intermediary that takes requests from the Angular web screens, fetches the correct legal learning data from the server, and delivers it back to the user.
- CDC: Centers for Disease Control and Prevention.
- HTML5 / Bootstrap / JavaScript: The presentation layer tech stack used for building the mobile-friendly web interface.
- Java / JDBC: The application layer tech stack used to power the scenario, authentication, and curation engines.
- JDL (JHipster Domain Language): Formatting language used to outline an application's database structure in a single file.
- JWT (JSON Web Token): A secure, encrypted digital badge used to track user logins.
- KIP: Knowledge Is Power digital legal literacy platform.
- MFCD: Major Functional Component Diagram detailing system architecture layout.
- PostgreSQL / MySQL: The relational database management systems utilized by KIP. The development environment leverages MySQL for local database management and testing, while the production layer utilizes PostgreSQL to ensure robust, enterprise-grade data persistence.
- REST (Representational State Transfer): An architectural style for designing networked applications that relies on a stateless, client-server communication protocol, typically HTTP.
- VPS (Virtual Private Server): A private, isolated section of a physical server acting as an independent computer.

5. References

- 5.1 Centers for Disease Control and Prevention. (2026). *Transportation Safety*.
<https://www.cdc.gov/transportation-safety/index.html>
- 5.2 Legal Services Corporation. (2017). *The justice gap: Measuring the unmet civil legal needs of low-income Americans*.
<https://www.lsc.gov/sites/default/files/images/TheJusticeGap-FullReport.pdf>
- 5.3 National Highway Traffic Safety Administration. (2024). *Teen driving: Talk to your teen driver about safe driving*.
<https://www.nhtsa.gov/teen-driving/parents-talk-your-teen-driver-about-safe-driving>
- 5.4 Touro Law Jacob D. Fuchsberg Law Center. (2017). *A Lawyer's Experience in K-12 Law-Related Education: Lessons and Opportunities*.
<https://digitalcommons.tourolaw.edu/cgi/viewcontent.cgi?article=1023&context=jel>
- 5.5 JHipster Team. (2026). *JHipster: A development platform to generate, develop and deploy Spring Boot + Angular / React / Vue Web applications and Spring microservices* (Version 8.x) [Computer software]. jhipster.tech

Listing of Figures

Figure 1: KIP Major Functional Component Diagram (MFCD)



Listing of Tables

Table 1: Presentation Layer

Layer	Component Type	Technologies / Specifications	Description / Functions
Presentation Layer	Software (Frontend)	JavaScript, TypeScript, Angular, Bootstrap, HTML5	Account Portal: Manages user registration, secure login validation, and restricted guest access functionality. Module Browser: Visualizes instructional legal domains (e.g., traffic law) and real-time progress statuses. Game Browser: Displays active games and surfaces tailored dashboard recommendations from the Curation System. Scenario Player: Primary gameplay interface displaying interactive scenes, branching options, tooltips, and legal source text.

Table 2: Application Layer

Layer	Component Type	Technologies / Specifications	Description / Functions
Application Layer	Software (Backend)	Java, Spring Boot, Spring Security, Spring Data JPA, REST APIs	Authentication System: Verifies active user sessions, tokens, credentials, and progressive feature permissions using stateless JWTs. Scenario Engine: Directs interactive matches, sequentially serving distinct narrative choices, structural forks, and game states. Feedback System: Delivers immediate, post-decision legal explanations tailored to student answers with plain-language reasoning. Curation System: Adapts and ranks module content based on historical user strengths and weaknesses. Source Reference System: Links scenarios to formal legal statutes while tracking external legal updates and flagging outdated data.

Table 3: Data Layer

Layer	Component Type	Technologies / Specifications	Description / Functions
Data Layer	Software (Database)	MySQL, PostgreSQL, Spring Data JPA, Liquibase	Database Units: Hosts secure relational tables for User, Progress, LearningModule, GameScenario, and LegalResource entities. Core Operations: Logs encrypted credentials, user profiles, quantitative progress logs, branching transition tables, and analytical metrics.

Table 4: Hardware Layer

Layer	Component Type	Technologies / Specifications	Description / Functions
Hardware Layer	Infrastructure	Virtual Private Server (VPS), Cloud Compute Nodes, Docker Compose	Server-Side Components: Hosts containerized backend application images and PostgreSQL services inside isolated Docker containers. Client-Side Components: Runs the responsive web interface smoothly inside standard consumer web browsers across smartphones, tablets, and desktops. Core Functions: Executes Java logic, manages volume data mounts, handles gateway traffic routing, and handles horizontal container scaling.